# Prática de Física dos Dispositivos Eletrônicos

## FGA0100

# Laboratório-5

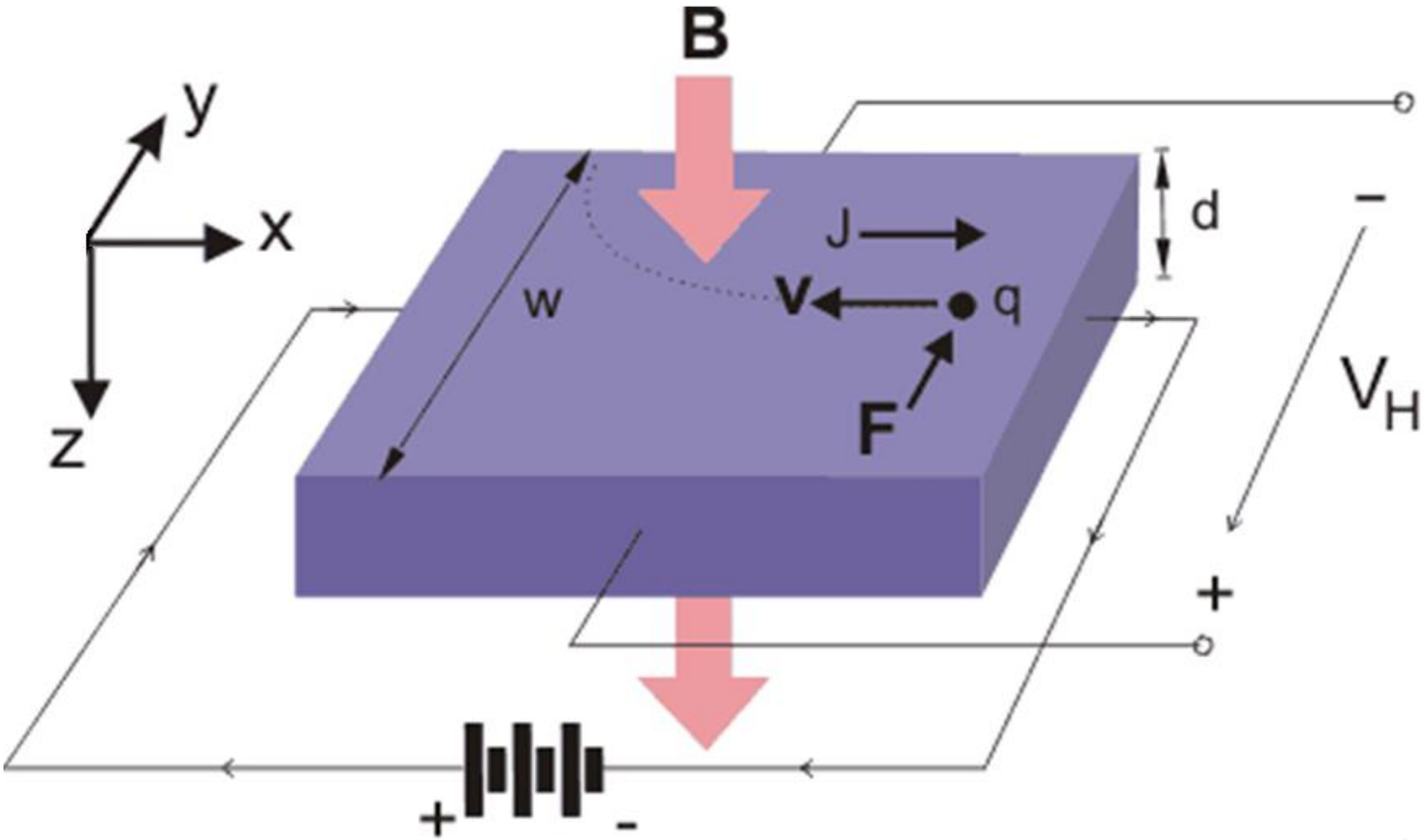
# Sensor Hall

# 01/2022

FGA

Universidade de Brasília

# Efeito Hall



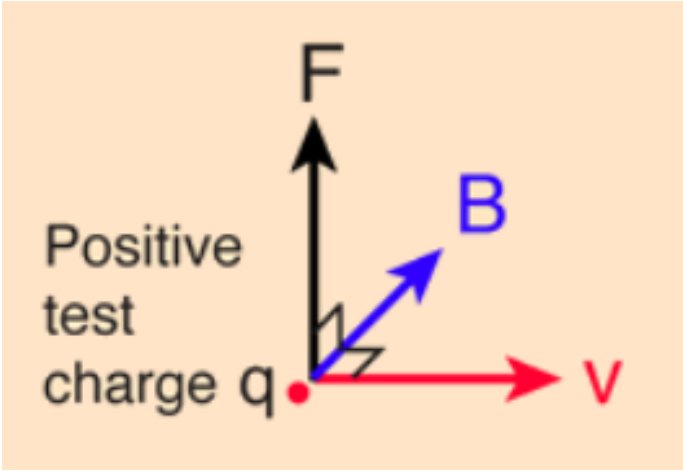
# Força de Lorentz

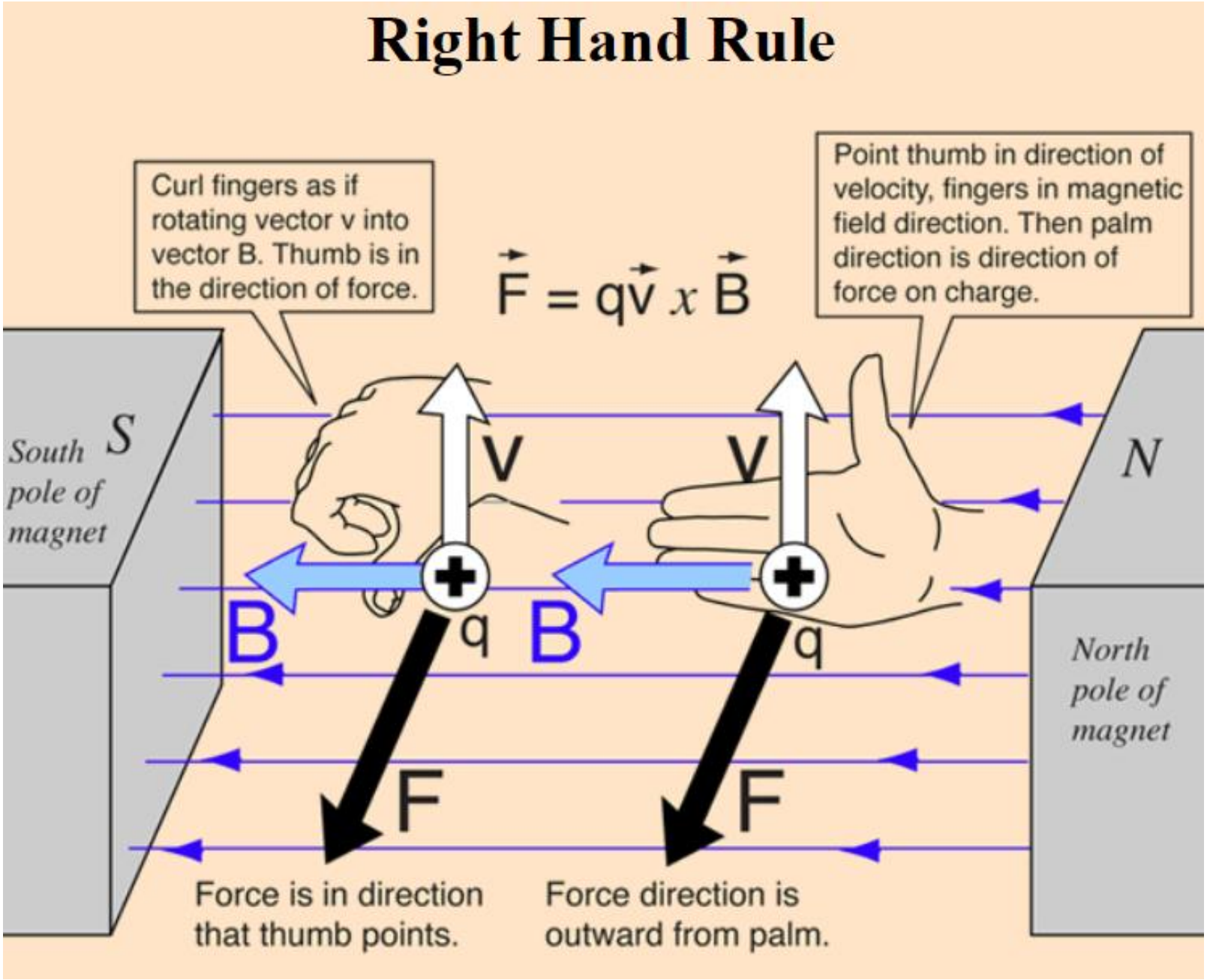
$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$F_x = q(E_x + v_y B_z - v_z B_y)$$

$$F_y = q(E_y + v_z B_x - v_x B_z)$$

$$F_z = q(E_z + v_x B_y - v_y B_x)$$



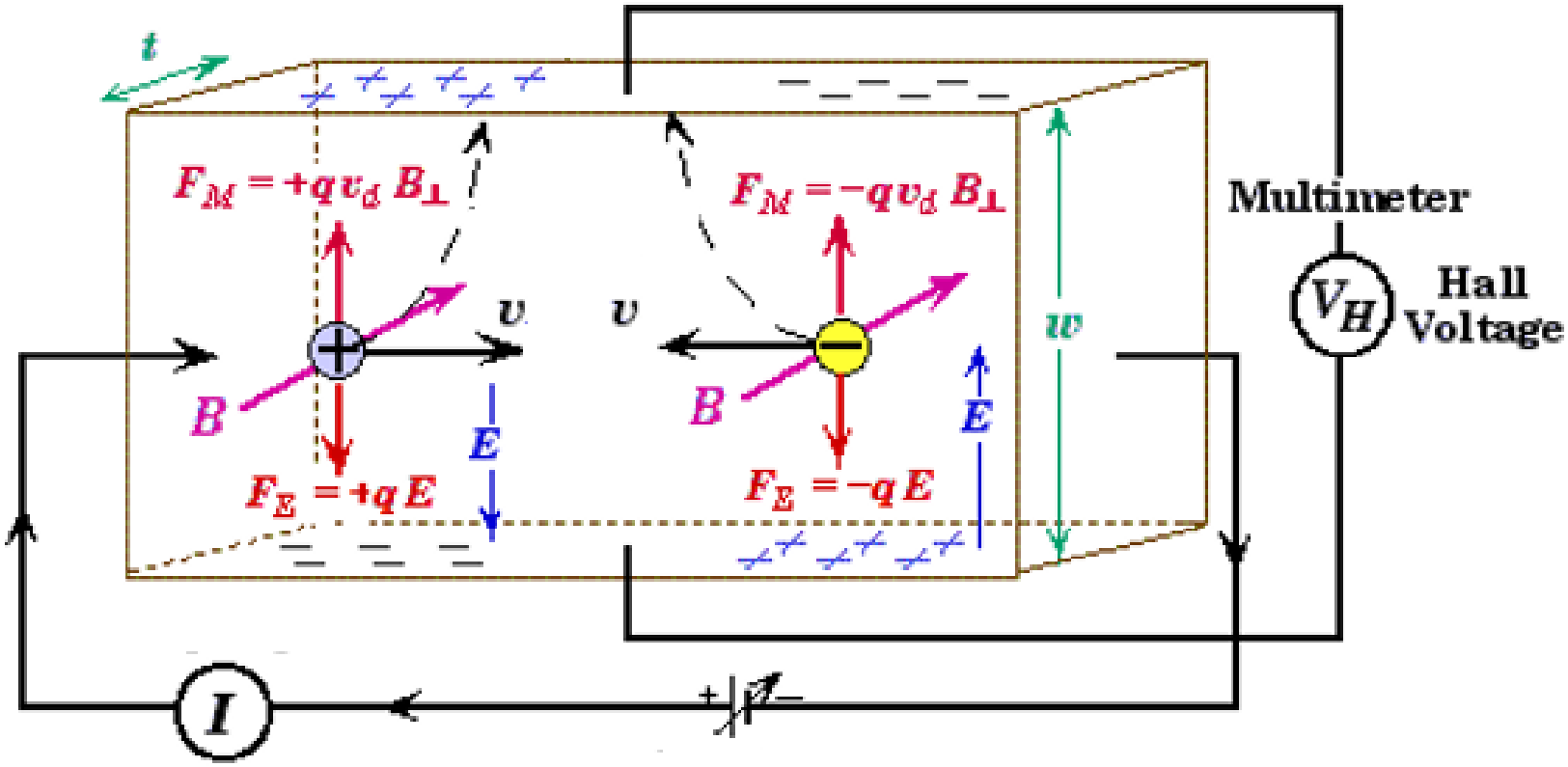



http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/magfor.html

# Efeito Hall

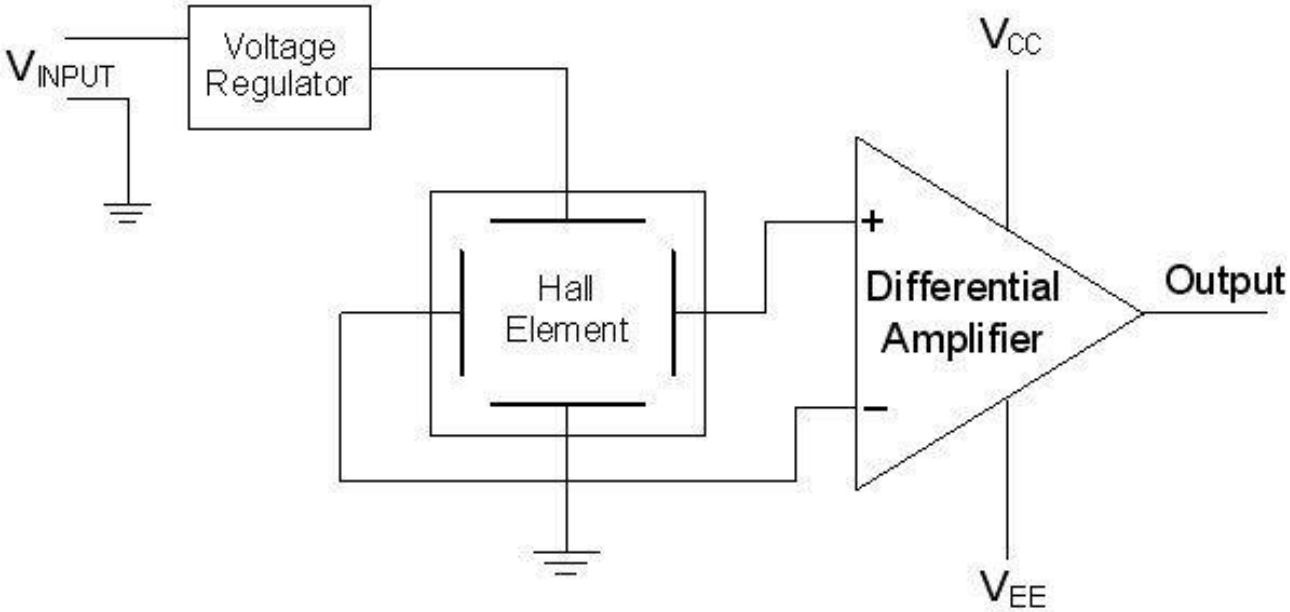
$$p_0 + N_D^+ = n_0 + N_A^-$$

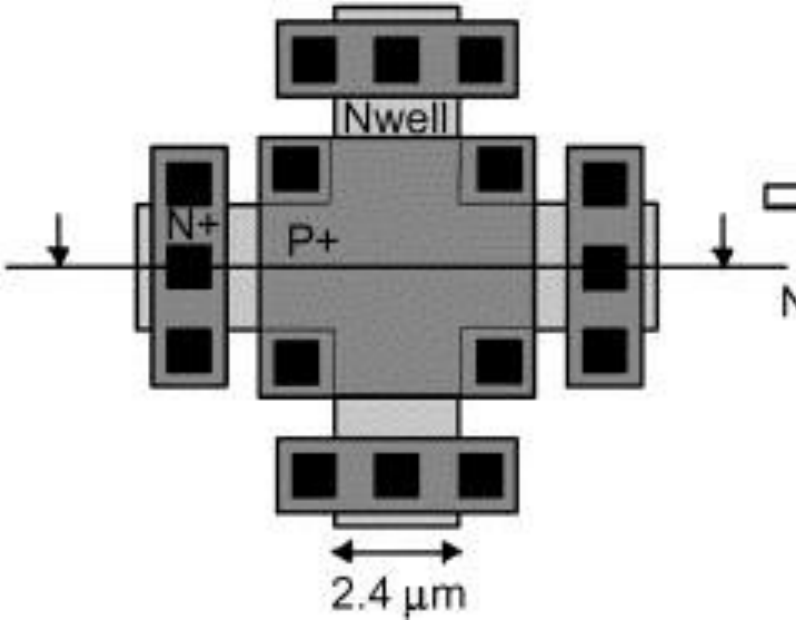
$$V_H = R_H J_P B_z$$

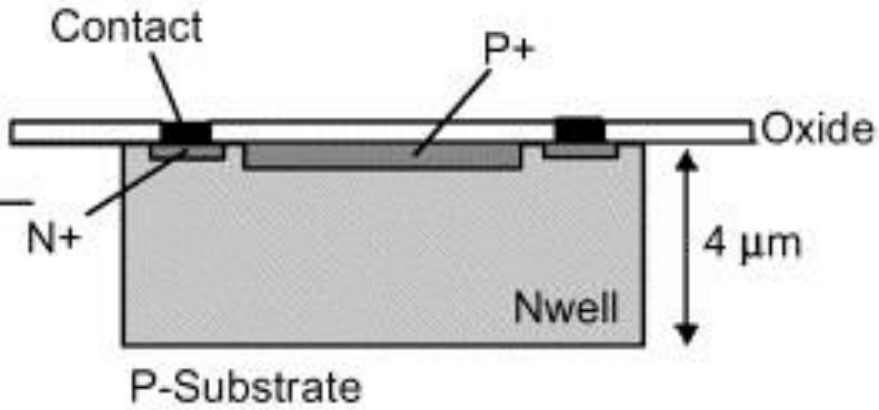



$$R_H = \frac{p\mu_h^2 - n\mu_e^2}{e(p\mu_h + n\mu_e)^2}$$

# Sensor Hall Integrado



# Densidade de Fluxo Magnético Terrestre

Intensidade: 25 to 65 µT (0.25 to 0.65 G)

1 tesla = 10.000 gauss

$$\mathrm{T} = \mathrm{V{\cdot}s{\cdot}m^{-2}} = \mathrm{N{\cdot}A^{-1}{\cdot}m^{-1}} = \mathrm{Wb{\cdot}m^{-2}} = \mathrm{kg{\cdot}C^{-1}{\cdot}s^{-1}} = \mathrm{kg{\cdot}A^{-1}{\cdot}s^{-2}} = \mathrm{N{\cdot}s{\cdot}C^{-1}{\cdot}m^{-1}}$$

| | | |
|---|---|---|
| A = ampere | m = meter | T = tesla |
| C = coulomb | N = newton | V = volt |
| kg = kilogram | s = second | Wb = weber |

31.869 µT ($3.1 \times 10^{-5}$ T) – strength of Earth's magnetic field at 0° latitude, 0° longitude

5 mT – the strength of a typical refrigerator magnet

0.3 T – the strength of solar sunspots

1.25 T – magnetic field intensity at the surface of a neodymium magnet

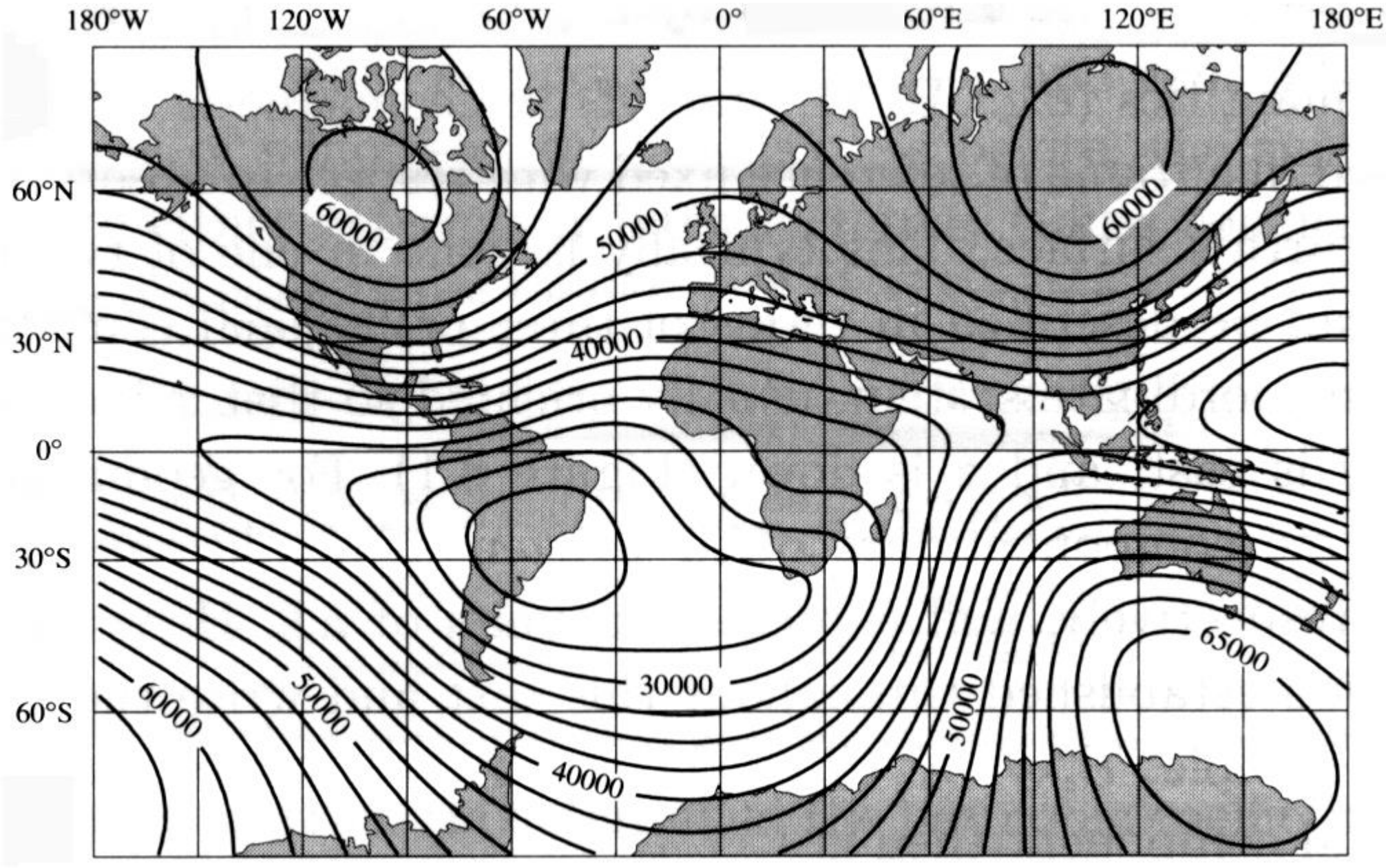
1 T to 2.4 T – coil gap of a typical loudspeaker magnet

1.5 T to 3 T – strength of medical magnetic resonance imaging systems in practice, experimentally up to 17 T

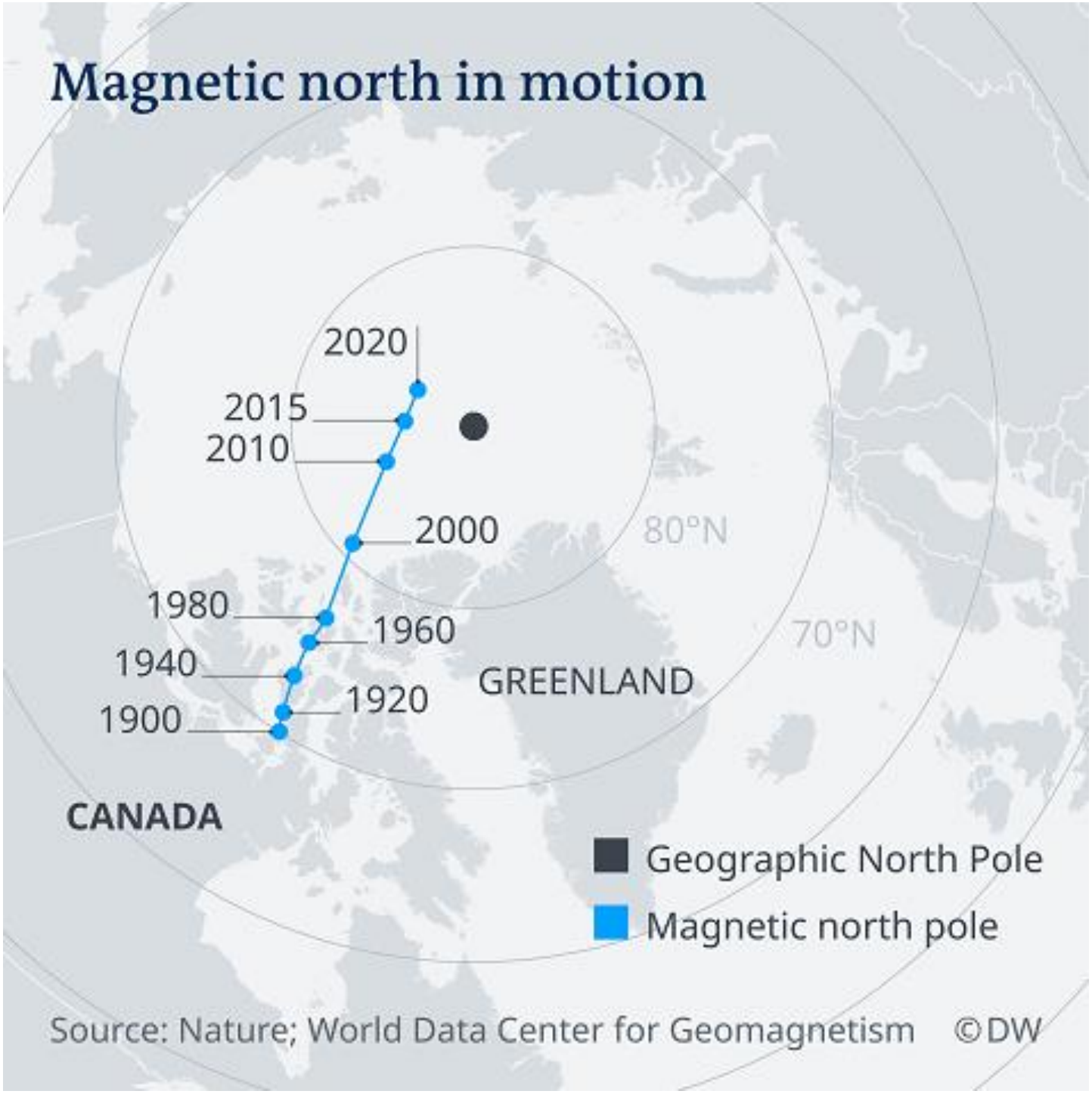
4 T – strength of the superconducting magnet built around the CMS detector at CERN

8 T – the strength of LHC magnets.

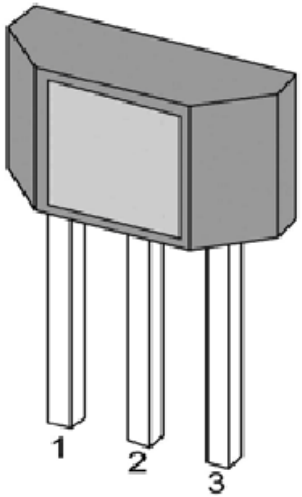
# Campo Magnético Terrestre



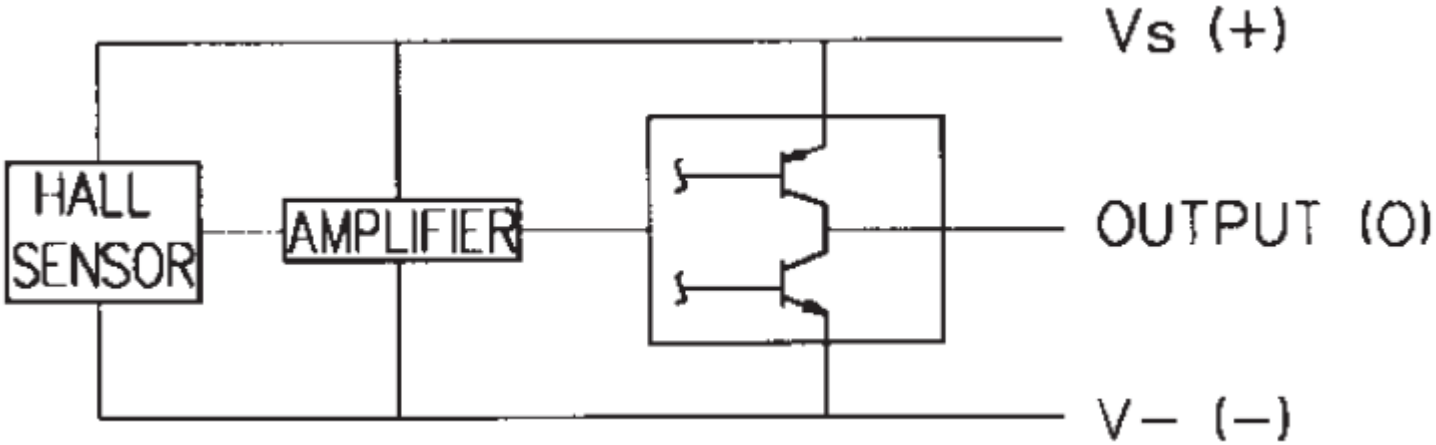

Intensidade em nT (nanotesla)


Magnetic north in motion
2020
2015
2010
2000
1980
1960
1940
1920
1900
80°N
70°N
GREENLAND
CANADA
Geographic North Pole
Magnetic north pole
Source: Nature; World Data Center for Geomagnetism
©DW

# Sensor Hall (SS495A)



| Name | No | Status | Description |
|---|---|---|---|
| Vdd | 1 | P | Power Supply |
| Gnd | 2 | P | IC Ground |
| Output | 3 | O | Output |



# Sensor Hall (SS495A)

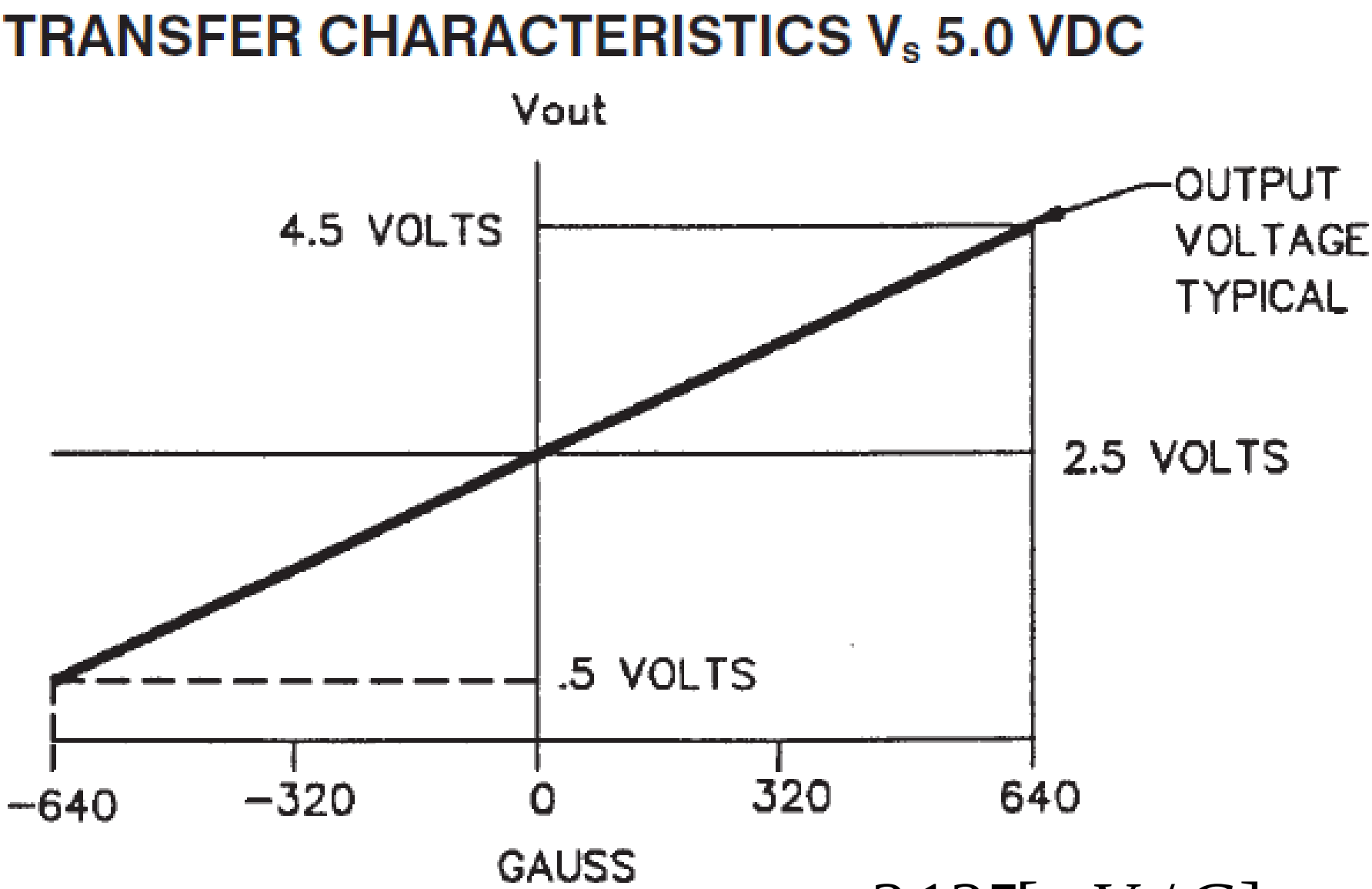
## Electrical Characteristics（TA =25℃， VCC =5.0V）

| Parameter | Symbol | Test Conditions | Min | Typ | Max | Units |
|---|---|---|---|---|---|---|
| Operating Voltage | $V_{CC}$ | Operating | 4.5 | 5 | 10.5 | V |
| Supply Current | $I_{CC}$ | Average | | 5 | 8.0 | mA |
| Output Current | $I_{OUT}$ | | 1.0 | 1.5 | | mA |
| Response Time | Tack | | | 3 | | uS |
| Quiescent Output Voltage | Vo | B=0G | | 2.5 | | V |
| Sensitivity | △Vout | $T_A$=25℃ | 3.0 | 3.3 | 3.6 | mV/G |
| Min Output Voltage | | B=-700G | | 0.2 | | V |
| Max Output Voltage | | B=700G | | 4.8 | | V |

## Absolute Maximum Ratings

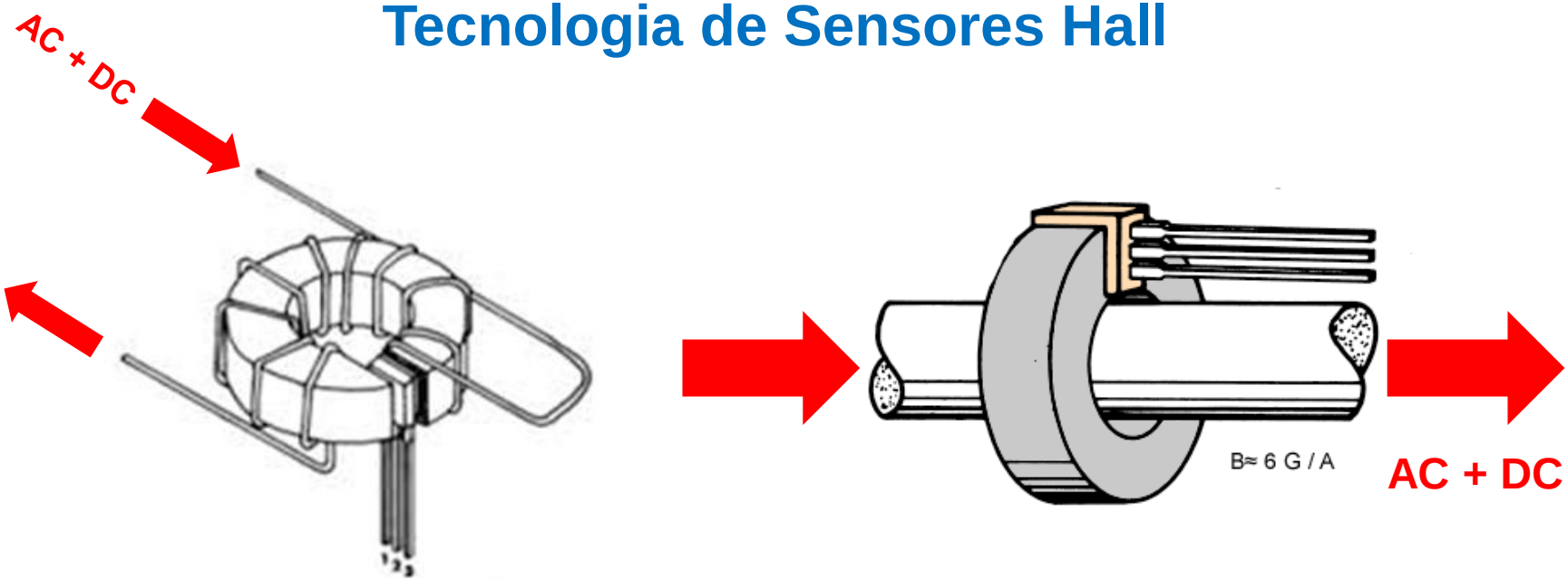
| Parameter | Symbol | Value | Units |
|---|---|---|---|
| Supply Voltage (operating) | $V_{CC}$ | 10.5 | V |
| Output Current | $I_{OUT}$ | 2 | mA |
| Operating Temperature Range | $T_A$ | -40~150 | ℃ |
| Storage Temperature Range | $T_S$ | -65~150 | ℃ |

# Sensor Hall (SS495A)

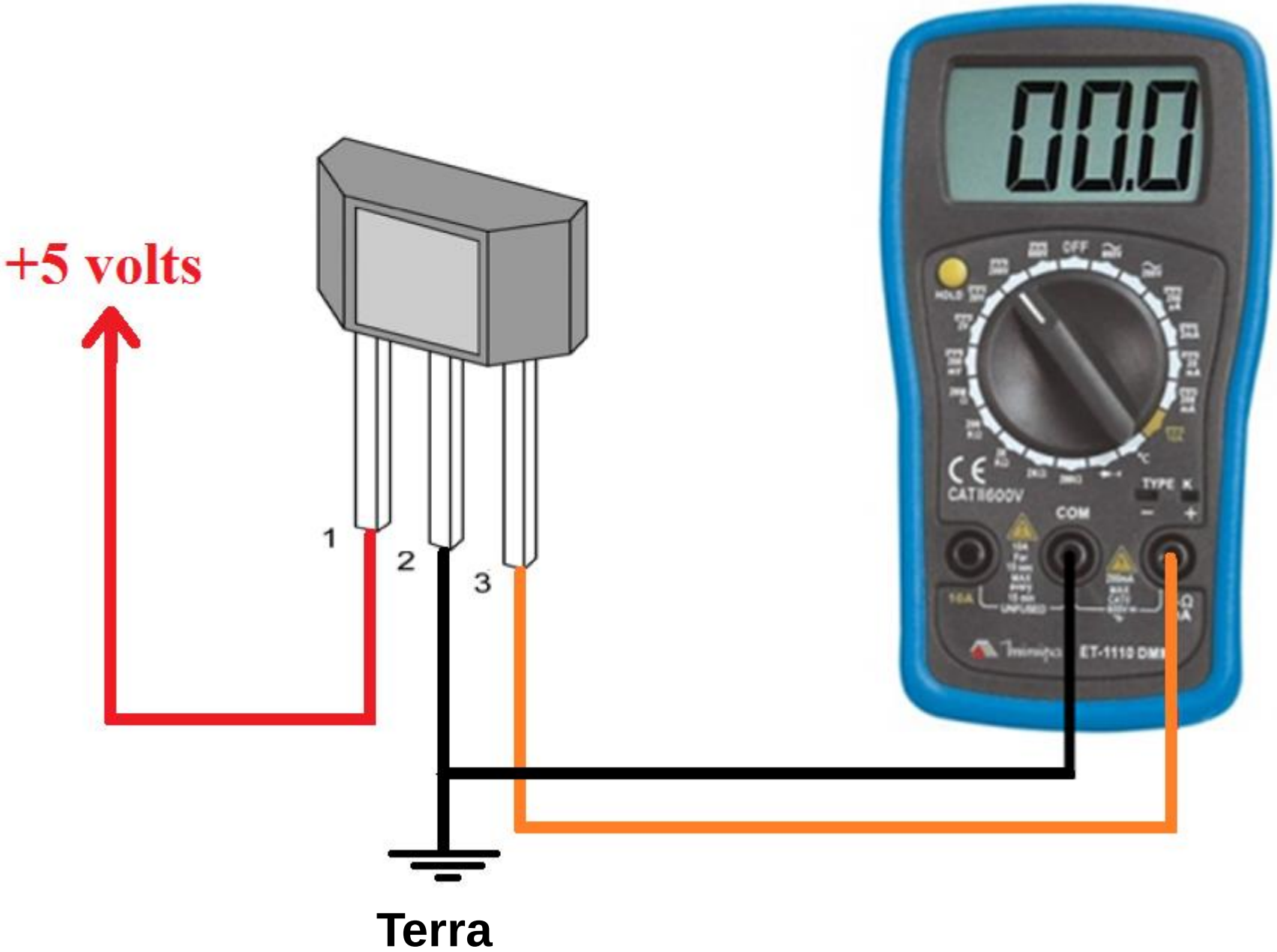



$$3{,}125[mV/G]$$

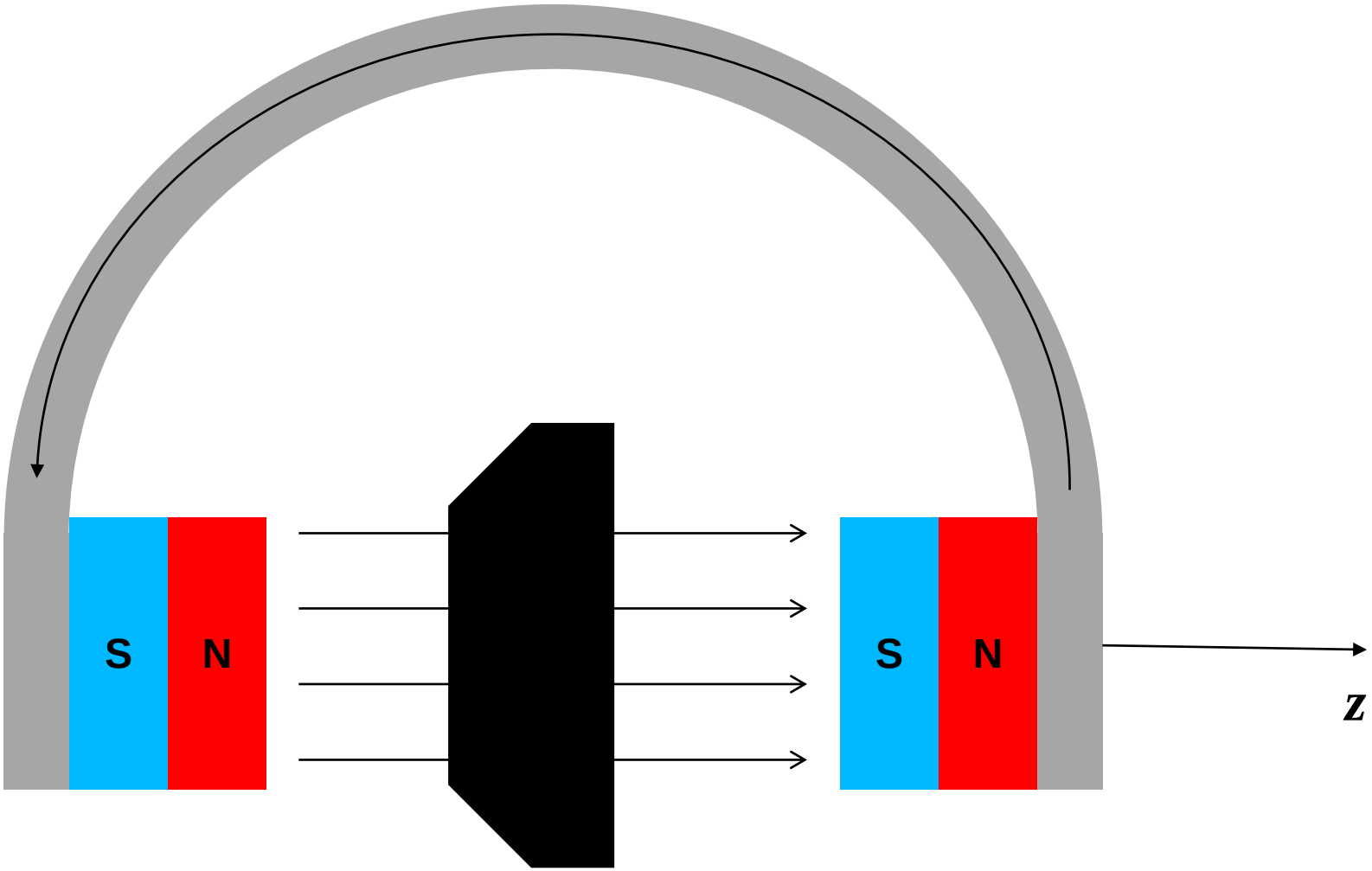
# Tecnologia de Sensores Hall



# Circuito de Medidas

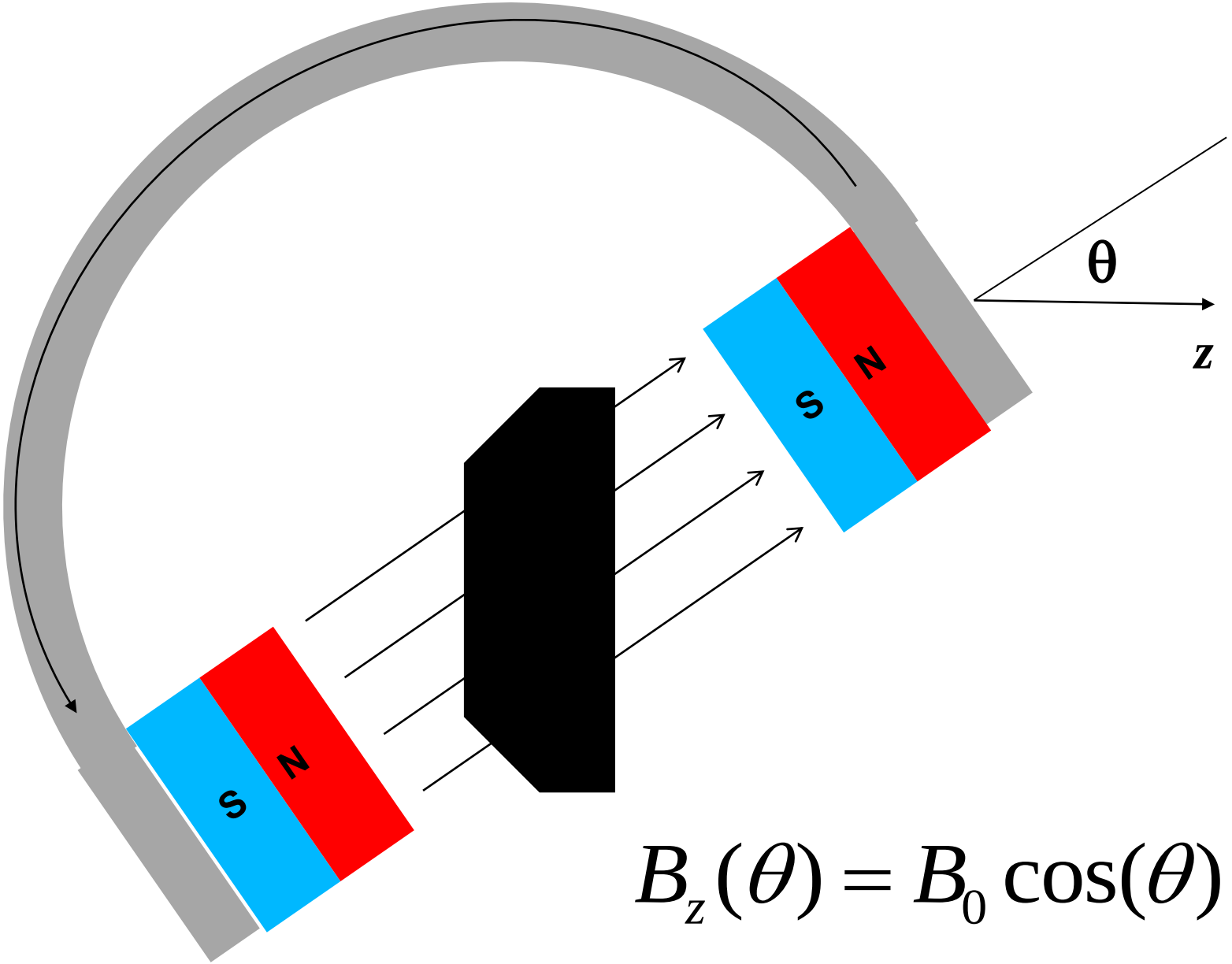


# Magneto



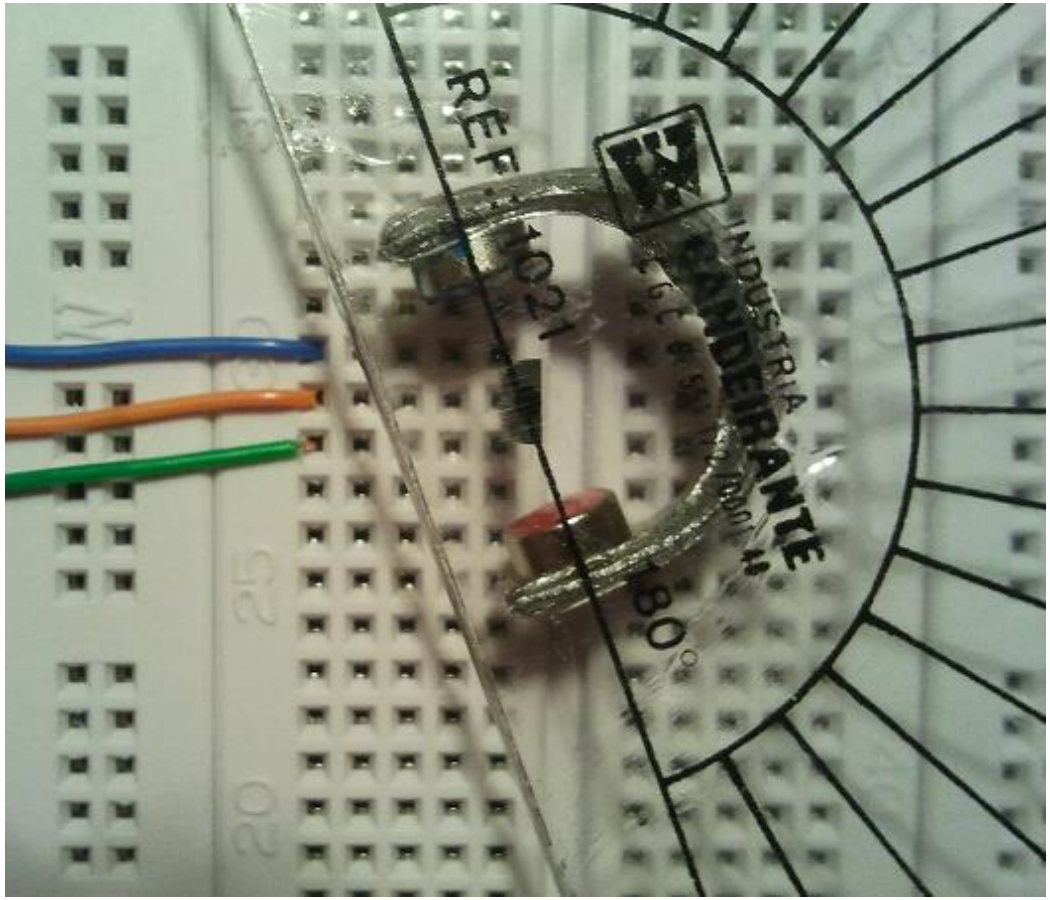

$$B_z = B_0$$

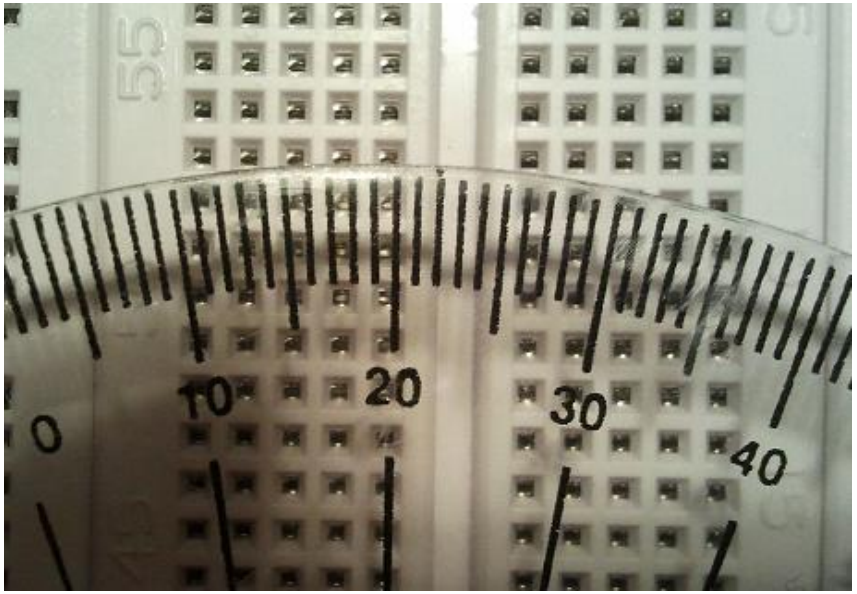
# Rotação do Magneto




$$B_z(\theta) = B_0 \cos(\theta)$$

# Medidas com o Sensor Hall



**Transferidor com Magneto em “U”**

# Modelo Físico

$$B_z(\theta) = B_0 \cos(\theta)$$

$$V_H = \frac{IB_z(\theta)}{ep_0 A} = \frac{J_P B_z(\theta)}{ep_0} = R_H J_P B_z(\theta) = k_H B_z(\theta)$$

**Tipo-p**

$$V_s = GV_H + V_{so}$$

$$V_{so} = V_s\Big|_{B_z=0}$$

$$V_s = Gk_H B_z(\theta) + V_{so} = Gk_H B_0 \cos(\theta) + V_{so}$$

$$\boxed{V_s = k_s B_0 \cos(\theta) + V_{so}}$$

## Determinando $B_0$

$$\boxed{V_s = k_s B_0 \cos(\theta) + V_{so}}$$

$$\theta = 0 \Rightarrow V_{s_\max} = k_s B_0 + V_{so}$$

$$\boxed{B_0 = \frac{V_{s_\max} - V_{so}}{k_s}}$$

$$k_s \cong 3{,}125 [mV / G]$$